

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 9: Transonic Design Features: Swept Wings & Area Rule

Syllabus Source: CAE Oxford (pp. 305-344) | Pass Target: >= 90% AviationExam

1. WING SWEEPBACK: THE EFFECTIVE VELOCITY VECTOR

Sweeping the wing rearward (sweep angle Lambda, typically 25° to 35°) is the primary design technique to increase Mcrit and Mdd:

- Velocity Decomposition: The oncoming freestream airspeed (V) is resolved into two orthogonal components: 1. Normal Component ($V_n = V \times \cos(\text{Lambda})$), perpendicular to the leading edge; 2. Spanwise Component ($V_s = V \times \sin(\text{Lambda})$), parallel to the leading edge.
- The Sweep Benefit: ONLY THE NORMAL COMPONENT (V_n) produces aerodynamic pressure changes and accelerates over the airfoil! The spanwise component produces only minor skin friction. Because $V_n = V \times \cos(\text{Lambda})$ is significantly smaller than V, the effective Mach number sensed by the wing is REDUCED: $M_{\text{effective}} = M \times \cos(\text{Lambda})$!
- Result on Mcrit: Sweeping a wing increases the aircraft's Critical Mach Number: $M_{\text{crit(swept)}} = M_{\text{crit(straight)}} / \cos(\text{Lambda})$. Allows the aircraft to cruise at much higher speeds before shock waves form!

2. SWEPT WING AERODYNAMIC DISADVANTAGES & REMEDIES

While wing sweep raises cruising Mach number, it introduces severe low-speed and stability penalties:

Swept Wing Penalty / Problem	Aerodynamic Physical Cause	Engineering Solution & Cockpit Remedy
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Wingtip Stall Tendency	Spanwise airflow boundary layer drift thickens at tips, causing wingtips to stall FIRST.	Wing washout (twist), leading-edge stall fences, vortilons, and vortex generators.
Severe Pitch-Up at Stall	Tip stall occurs behind CG on swept wings; loss of tip lift shifts resultant lift FORWARD!	Stick Pusher system; low-mounted horizontal stabilizer to avoid deep stall wash.
Dutch Roll Instability	Strong dihedral effect of swept wings produces coupled yaw-roll oscillation.	Yaw Damper system (mandatory for flight dispatch under MEL).
Reduced C _L _max & High Stall Speed	Swept wings have a flatter lift curve slope ($dC_L/d\alpha$ is lower).	Complex multi-slotted Fowler flaps and leading-edge slats mandatory for take-off/landing.

3. SUPERCRITICAL AIRFOILS & THE WHITCOMB AREA RULE

- Supercritical Airfoil: Features a FLATTENED upper surface (limits peak airflow acceleration, delaying and weakening shock waves), high aft camber (restores lift near the trailing edge), and a well-rounded leading edge. Allows higher cruising Mach numbers or thicker wings (saving structural weight).
- Whitcomb Area Rule: To minimize transonic wave drag, the cross-sectional area of the entire aircraft (fuselage + wings + engines) plotted along the longitudinal axis must change SMOOTHLY without abrupt steps. Achieved by 'pinching' the fuselage at the wing root (coke-bottle fuselage waist) or adding flap track canoe fairings / Küchemann carrots.

- Vortex Generators: Small, low-aspect-ratio vertical blades mounted on the wing upper surface at an angle of attack. Shed high-energy vortices that mix freestream energy into the boundary layer, preventing shock-induced flow separation.

[!] EXAM TRAP (AVIATIONEXAM) - Effective Mach Number on a Swept Wing Calculation

- Formula: $M_{\text{effective}} = M_{\text{free}} \times \cos(\text{Sweep Angle } \Lambda)$! - Example: An aircraft with 30° wing sweep flies at Mach 0.80. - The wing airfoil behaves as if it were flying at: $M_{\text{eff}} = 0.80 \times \cos(30^\circ) = 0.80 \times 0.866 = \text{MACH } 0.69$! - This delays shock wave formation by over 0.10 Mach!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Swept Wing Effective Mach Calculation

Problem: An airliner flies at Mach 0.80 with 30 degrees wing sweep ($\cos 30 = 0.866$). What Mach number does the airfoil aerodynamically experience perpendicular to the leading edge? Step 1: Formula: $M_{\text{effective}} = M_{\text{flight}} \times \cos(\text{Sweep})$. Step 2: $M_{\text{effective}} = 0.80 \times 0.866 = \text{Mach } 0.69$. Conclusion: The aircraft flies at Mach 0.80 while its wing profile experiences only Mach 0.69, delaying wave drag.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Why does sweeping an aircraft wing rearward increase the Critical Mach Number (M_{crit})?

- [A]: It increases wing aspect ratio
- [B]: It decomposes oncoming airflow, so that only the component PERPENDICULAR to the leading edge ($V \times \cos(\Lambda)$) determines aerodynamic pressure acceleration
- [C]: It eliminates induced drag
- [D]: It shifts the CG forward

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The chordwise pressure distribution on a swept wing depends solely on the velocity component normal to the leading edge ($V_n = V \cos(\text{sweep})$). Because V_n is lower than true airspeed, M_{crit} is delayed to a higher speed.

Q2: What dangerous aerodynamic stalling characteristic is inherently associated with a sweptback wing?

- [A]: Wing root stall with severe nose-down pitch
- [B]: WINGTIP STALL first, which causes loss of aileron roll control and an uncommanded violent NOSE-UP pitching moment
- [C]: Inability to stall
- [D]: Immediate flameout

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Swept wings suffer spanwise boundary layer flow that thickens the tip boundary layer, causing wingtips to stall first. Because swept wingtips are located well behind the CG, losing tip lift causes a strong pitch-up.

Q3: What design feature characterizes a 'Supercritical' airfoil?

- [A]: A highly cambered upper surface

- [B]: A flattened upper surface to weaken shock waves, combined with high aft camber to generate lift
- [C]: A sharp knife-like leading edge
- [D]: Negative sweep

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Supercritical airfoils feature a flattened upper surface that delays local supersonic acceleration and weakens shock waves, while deep underside curvature near the trailing edge restores required lift.